

- TP 1 : Introduction to Cloud
- Professor: Ouassim Karrakchou
- Level : 4th year

TP 1 Part 2

Introduction to Cloud

Objectives:

The goal of this TP is to introduce you to the core concepts of the Cloud

We will do the following tasks:

- Get your access to the AWS Academy and Azure for Students accounts.
- Discover the key architectural components of a Cloud (regions, availability zones, services).
- Discover the basic compute and storage services available in Azure and AWS.

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
 - a. Awatif ROUIJEL awatifrouijel@gmail.com
 - b. Mohammed SOUIDI souididoc@gmail.com
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

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I- Introduction:

The first TP of the Cloud Foundations course will introduce you to the Cloud. You will first get your student access to the Microsoft Azure and Amazon AWS clouds, and follow an e-learning on the official AWS academy platform that introduces you to the Cloud, as well as provide with several labs where you will discover how to configure the AWS Cloud to create several resources. In the first part, you learned how to create a compute resource in the Cloud: a virtual machine. Today, we will focus on learning how to create another type of Cloud resource: storage resources. You will see 3 types of storage resources: block storage (that behaves like a virtual hard drive or SSD), object storage (that can store files and objects online), and databases (that can store data in an organized manner).

II- Create a block storage in AWS:

1. Connect to the **AWS Academy portal**, and access the **Cloud Foundations course**.
2. Go the **Module 7**
3. Click on Lab 4, then on Load Lab 4 to open the lab window.
4. Follow the instructions in the lab to create your Elastic Block Storage (EBS) in AWS and attach it to a VM. An EBS is a virtual hard drive or SSD that can be accessed by compute services like VMs. It can be formatted with any file system you want.
5. There are 6 tasks in the lab. Whenever you finish a task (except for task 5), **call the Professor** to show him/her what you did.

IV- Create a Blob storage in Azure and host a static website in it:

1. Go now to the following page : <https://learn.microsoft.com/en-us/training/modules/guided-project-deploy-static-website-blob-storage/>
2. Follow the instructions in the guided project to create your **first Blob storage** in Azure and use it to host a **static website**.
3. There are 3 exercises in the lab. Whenever you finish an exercise, **call the Professor** to show him/her what you did.

V- Create a simple SQL database in Azure:

1. Go now to **the Azure Portal**
2. Your goal is to create a **simple SQL database** in Azure, create a simple **table** called **STUDENTS** with 2 columns : **family_name** and **first_name**, and **insert your name** in it using an **SQL query** in Azure.

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3. Investigate how to do this by **searching online on the Azure documentation**. Then **do it in your Azure portal**.
4. **Call the Professor** to show him/her what you did.
5. **Delete the SQL database and all the associated resources** to not consume credits from your Azure for Students account

VI- Assignment:

Based on what you learned in the TP explain the different steps needed to create the 3 types of storage resources you have seen today in Azure and in AWS. Use screenshots from the Azure portal and the AWS console. **For the case of Azure, when you have finished, make sure you delete all the resources you used (show the steps).**